

Neuronal Avalanches as a predictive biomarker of stroke patients

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Abstract—Stroke affects about 15 million people worldwide each year, causing 5 million deaths and leaving another 5 million permanently disabled [1]. Given the limited motor recovery observed in most patients, the TiMeS project aims to better understand stroke-related brain mechanisms and identify predictive biomarkers of recovery [2].

Within this framework, this project evaluated the predictive power of EEG-derived features using multiple binary machine learning classifiers to distinguish stroke patients from healthy controls. In addition, SHAP analysis was applied to improve model interpretability and identify the brain regions most relevant for stroke classification [3].

I. INTRODUCTION

Artificial Intelligence (AI) is increasingly used to support clinical diagnosis and prognosis. While AI has shown effectiveness with stroke imaging data [4], stroke-related EEG remains underexplored despite its potential to inform post-stroke recovery.

This project aims to classify stroke patients and healthy old subjects using resting-state EEG features, comparing the predictive power of Coherence matrix (Coh) and Avalanche Transition Matrix (ATM). The hypothesis is that neuronal avalanches (aperiodic bursts of brain activity) better capture pathological dynamics than coherence, which assumes stationary signals, supported by evidence of their predictive power in epilepsy [5].

Employed stroke data derive from the TiMeS dataset, a multimodal longitudinal study [2]. Only EEG from the first time-point (T1, one week post-stroke) was used, acquired in resting state with a 64-channel system and preprocessed for artifact removal, epoching, and source reconstruction [2]. The dataset included 54 stroke patients and 14 healthy old subjects as controls.

II. METHODS: FEATURE EXTRACTION

A. Coherence matrix

Coherence is a statistical metric that measures the correlation between two signals in a specific frequency range. It is defined as the ratio of the cross-spectrum of x and y to the product of auto-spectrum of x and y :

$$C_{xy} = \frac{|P_{xy}(f)|^2}{|P_{xx}(f)||P_{yy}(f)|} \quad (1)$$

Coherence represents a functional connectivity measure among brain regions in the frequency domain, and can be used to study functional interplay of different brain areas in certain pathological settings. However, for the computation of power

spectrum density (PSD) and cross-spectrum density (CSD), wide-sense stationarity of the process is required, even though this assumption might not reflect the reality of brain networks dynamics, often characterized by bursts and transients.

EEG epochs for each patient were loaded into MNE objects [6], and coherence matrices were computed using *spectral_connectivity_time* in the two typical resting-state frequency bands: α [8–12 Hz] (rest) and β [12–30 Hz] (awake/alert).

B. ATM matrix

Avalanche Transition Matrix is a metric extracted from Neuronal Avalanches (NA), defined as events starting when the z-scored EEG signals cross a specific threshold in any region and ending when all regions return to baseline [5]. For each subject, EEG epochs were concatenated, z-scored over time, thresholded at 2.5, and binarized, setting over-threshold points to 1 and others to 0 [7]. Consequently, the signal was binned to capture the statistical-mechanical properties that NA exhibit, which are theoretically associated with a branching process operating close to a critical regime. Since selecting an appropriate time-bin allows to capture critical brain dynamics [5], the branching factor was computed for each detected NA (σ_i) and the time-bin size optimized in order to obtain an overall branching factor $\sigma \sim 1$ (bin size = 1).

$$\sigma_i = \prod_{j=1}^{N_{bin}-1} \left(\frac{n_{events}(j+1)}{n_{events}(j)} \right)^{\frac{1}{N_{bin}-1}} \quad (2)$$

$$\sigma = \prod_{i=1}^{N_{aval}} \sigma_i^{\frac{1}{N_{aval}}} \quad (3)$$

Time-bins with at least one over-threshold region were labeled active, and NAs of user-defined minimum duration were extracted. For each NA, related features such as size (number of active regions) and duration (in time-bins) were also computed. To assess features impact, statistical comparisons were made between acute stroke and healthy old groups on avalanches size and duration (mean and maximum), and branching factor. Swarm plots provided an initial qualitative view, followed by group-wise descriptive statistics. Normality and variance tests determined whether to use a two-sample t-test or Mann-Whitney U test, with Cohen's d and AUC quantifying effect size and discriminative power. FDR correction controlled for multiple comparisons. Since none of the features showed meaningful predictive value, they were excluded from the final

models.

Finally, the ATM matrix was computed for each detected NA and the resulting subject-level ATM obtained as the average over single-avalanche ATMs. ATM represents in the element (i, j) the probability that region j is active at time $t+1$, given that region i was active at time t : $ATM(i, j) = Pr(A|B) = \frac{Pr(A \cap B)}{Pr(B)}$, with event $A = \text{"region } j \text{ active at time } t+1\text{"}$ and $B = \text{"region } i \text{ active at time } t\text{"}$. The probability that region i is active at time t and that region j is active at $t+1$ is given by the number of transitions of activity that occur from region i to region j in subsequent time-points over the total number of transitions that may occur in subsequent time-points from i to j , i.e. the number of time-points

$$Pr(A \cap B) = \frac{\#(i \rightarrow j)_{t \rightarrow t+1}}{\#times} \quad (4)$$

The probability that region i is active at time t is given by the number of time-points where i is active over the total number of time-points.

$$Pr(B) = \frac{\#times \text{ active } i}{\#times} \quad (5)$$

Thus:

$$ATM(i, j) = Pr(A|B) = \frac{\#(i \rightarrow j)_{t \rightarrow t+1}}{\#times \text{ active } i} \quad (6)$$

III. METHODS: CLASSIFICATION

Binary classification was performed on the flattened features using three algorithms: Support Vector Machine (SVM), Random Forest (RF), and XGBoost. SVM was used to assess whether the feature representations were separable in a high-dimensional space. RF was included to investigate potential non-linear relationships and feature interactions. XGBoost was evaluated to determine whether more complex, boosted decision boundaries could uncover subtle connectivity patterns associated with stroke pathology. By comparing performance across these diverse models, the aim was also to determine whether observed classification trends were consistent and reflective of the underlying feature representations, rather than being driven by a specific classifier.

For each model, features and labels (1 = stroke, 0 = healthy) were loaded from a DataFrame. Data were split 80/20 for training and testing, stratified by label. To address class imbalance (54 vs. 14), the majority class was under-sampled and the minority class over-sampled using SMOTE [8] (applied only to the training set to preserve the test set as ground truth).

Data were then z-scored and a 3-fold cross validation set up to find the best hyperparameters combination for each model. In particular, the following combinations were tested:

A. Random Forest

- *rf_n_estimators*: [100, 300, 500] (number of trees)
- *rf_max_depth*: [None, 10, 20, 40] (tree depth)
- *rf_min_samples_split*: [2, 5, 10] (when to split a node)
- *rf_min_samples_leaf*: [1, 2, 4] (leaf size)
- *rf_max_features*: ['sqrt', 'log2'] (feature subsampling)

B. Support Vector Machine

- *svm_kernel*: ['linear', 'rbf']
- *svm_C*: [0.1, 1, 10, 100] (regularization strength)
- *svm_gamma*: ['scale', 'auto'] (only relevant for RBF)

C. XGBoost

- *xgb_max_depth*: [3, 5, 7] (tree complexity)
- *xgb_min_child_weight*: [1, 5, 10] (tree complexity)
- *xgb_learning_rate*: [0.01, 0.05, 0.1]
- *xgb_n_estimators*: [100, 300, 500]
- *xgb_subsample*: [0.6, 0.8, 1.0] (regularization)
- *gb_colsample_bytree*: [0.6, 0.8, 1.0] (regularization)

Both during cross validation and test time, the model was evaluated in terms of balanced accuracy (BA), to account for classes imbalance, expressed by the following formula $BA = \frac{1}{2} \cdot (\frac{TP}{TP+FN} + \frac{TN}{TN+FP})$

IV. METHODS: SHAP ANALYSIS

Understanding feature contributions improves model interpretability and supports clinical decision-making. Among interpretability methods, SHapley Additive exPlanations (SHAP) is a widely used feature-based approach grounded in game theory, where each feature is treated as a player and its Shapley value ϕ_j quantifies its contribution to the model prediction [3]. Shapley values are computed as a weighted average of a feature's marginal contribution across all possible feature coalitions:

$$\phi_j = \sum_{S \subset N \setminus \{j\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} (V(S \cup \{j\}) - V(S)) \quad (9)$$

Where S is a coalition, N is the set of all players, $(V(S \cup \{j\}) - V(S))$ is the marginal contribution of player j to the coalition S , $\frac{|S|!(|N| - |S| - 1)!}{|N|!}$ is the weight of the marginal contribution, $\sum_{S \subset N \setminus \{j\}}$ sums over all possible coalitions without player j . Shapley values can be interpreted as the weighted average of a certain player's marginal contribution across all possible subsets of players. In the interpretability perspective, for a given sample i , the SHAP values assigned to each feature quantify its contribution to the difference between the individual prediction and the average one.

SHAP methods depend on the model type. Since Kernel SHAP, compatible with SVM, has high computational cost we computed SHAP values using XGBoost instead, leveraging TreeExplainer to exploit the model's tree structure. In addition, model interpretability is important and more easily reachable with linear models such as SVM, but more difficult with non-linear models such as XGB. Hence, XGB-based SHAP was deemed more interesting.

Classification was repeated across 50 seeds to improve robustness. Due to SHAP Tree Explainer's dependence on model structure and the high feature-to-sample ratio (3844 features), feature attributions varied even with similar accuracies. Reliable interpretations were obtained by analyzing SHAP value distributions and their means (Fig. 6).

V. RESULTS: FEATURE EXTRACTION

A. Coherence

Visual result of the Coherence extraction is showed hereafter for one stroke patient under the α and β frequency band:

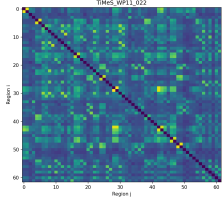


Fig. 1: α band

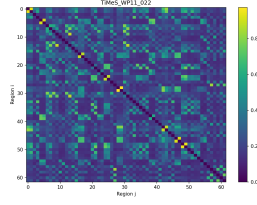


Fig. 2: β band

B. Scalar features of Neuronal Avalanches

Qualitative inspection (Fig. 3) and quantitative analysis (Table I) showed none of the scalar features was statistically significant.

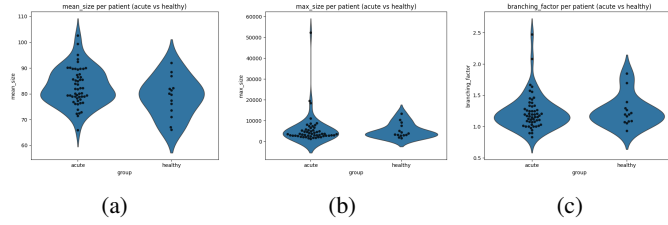


Fig. 3

TABLE I: Statistical analysis results for scalar features.

Feature	Test	p	AUC	p_{FDR}	Reject
mean_size	t-test	0.0789	0.6217	0.3946	False
max_size	MWU	0.6008	0.4537	0.7215	False
mean_duration	t-test	0.3489	0.5754	0.7215	False
max_duration	MWU	0.7215	0.5317	0.7215	False
branching_factor	MWU	0.6436	0.4590	0.7215	False

C. ATM

Visual results of the ATM extraction is showed hereafter for two stroke patients.

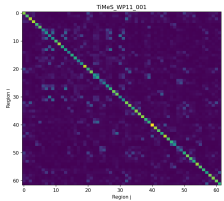


Fig. 4

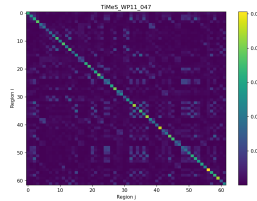


Fig. 5

VI. RESULTS: CLASSIFICATION

Once the three models were trained, the following balanced accuracies were obtained, by running the classifiers 50 times with different random states.

A. Coherence, ATM and hyperparameters

All the classifiers were run with Coh features extracted both in the α and β frequency bands.

TABLE II: Balanced accuracy of the evaluated models

Model	Balanced Accuracy (%)
SVM (α band)	$62,9 \pm 15,9$
RF (α band)	$55,3 \pm 13,0$
XGBoost (α band)	$53,1 \pm 13,1$
SVM (β band)	$67,3 \pm 15,0$
RF (β band)	$62,9 \pm 14,4$
XGBoost (β band)	$62,0 \pm 17,7$

Then, the 3 models were trained on ATMs with a bin size of 1, z-threshold of 2,5 [5] and 3 different minimum durations of the avalanche (20, 30, 40), leading to the following balanced accuracies.

TABLE III: Balanced accuracy of the evaluated models

Model	Balanced Accuracy (%)
SVM ($min_duration = 20$)	$74,2 \pm 13,2$
RF ($min_duration = 20$)	$64,8 \pm 13,2$
XGBoost ($min_duration = 20$)	$67,5 \pm 16,4$
SVM ($min_duration = 30$)	$74,5 \pm 13,5$
RF ($min_duration = 30$)	$68,9 \pm 15,3$
XGBoost ($min_duration = 30$)	$68,8 \pm 17,0$
SVM ($min_duration = 40$)	$73,3 \pm 14,9$
RF ($min_duration = 40$)	$67,4 \pm 15,2$
XGBoost ($min_duration = 40$)	$68,7 \pm 17,4$

Grid search identified the following most frequent best hyperparameters across 50 seeds:

- SVM: 'svm_C': 0.1, 'svm_gamma': 'scale', 'svm_kernel': 'linear'
- RF: 'rf_max_depth': None, 'rf_max_features': 'sqrt', 'rf_min_samples_leaf': 4, 'rf_min_samples_split': 2, 'rf_n_estimators': 100
- XGBoost: 'xgb_colsample_bytree': 0.6, 'xgb_learning_rate': 0.01, 'xgb_max_depth': 3, 'xgb_min_child_weight': 1, 'xgb_n_estimators': 500, 'xgb_subsample': 0.6

VII. RESULTS: SHAP ANALYSIS

Visualization of the distribution of SHAP analysis results

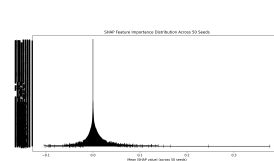


Fig. 6

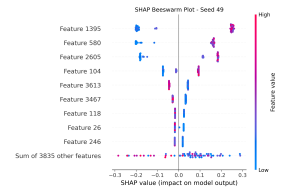


Fig. 7

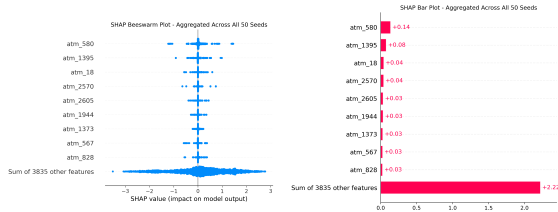


Fig. 8

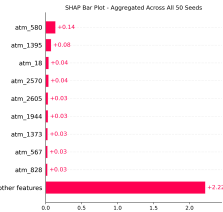


Fig. 9

TABLE IV: Top 5 Features Ranked by Mean Absolute SHAP Value (Connection Importance)

Feature	ROI_i	ROI_j	mean_abs_shap	std_abs_shap
atm_580	ROI_10	ROI_23	0.138184	0.240705
atm_1395	ROI_23	ROI_32	0.077033	0.170073
atm_18	ROI_1	ROI_19	0.042858	0.107857
atm_2570	ROI_42	ROI_29	0.0413	0.124932
atm_2605	ROI_43	ROI_2	0.033877	0.084831

VIII. DISCUSSION

A. ATM versus Coherence

As shown in Tables II and III, ATM features consistently outperformed coherence (baseline model) in classifying stroke patients versus healthy subjects across all classifiers, and had best performance with a minimum duration of 30 time bins. This result is particularly compelling and represents a scientific novelty as it aligns with the project’s initial hypothesis, which posited that neuronal avalanches, due to their complex and self-organizing nature, offer a more robust representation of brain activity compared to traditional connectivity metrics like coherence. The higher BA of ATM-based classifiers suggests that the dynamics of NA (spontaneous, scale-invariant activity patterns) observed in neural networks, are more sensitive to the subtle yet significant alterations in brain activity associated with stroke, that might not be captured by static measures such as coherence, which assumes stationarity of EEG recording.

B. Best model

A remarkable evidence from the balanced accuracies of the trained models is that their standard deviations result to be really large. However, it is worth reminding that the provided dataset was limited to 68 subjects, and only 14 ended up in the test set, leading to this statistical oscillation of the results. Across all minimum durations, SVM achieved the highest performance, peaking at a 30-sample (6 ms) minimum duration with hyperparameters ‘C’: 0.1, ‘gamma’: ‘scale’, ‘kernel’: ‘linear’.

SVM outperformed Random Forest and XGBoost in our dataset. This may be attributed to the high dimensionality of the feature space: SVMs are particularly effective in high-dimensional settings because they optimize a global separating hyperplane and are less prone to overfitting with many features, whereas tree-based methods may struggle to identify meaningful splits when many features are irrelevant or sparse, which is particularly true for ATM, as can be observed in Figures 4 and 5

C. SHAP analysis and interpretability

Figure 6 shows that most features have mean SHAP values close to zero, indicating low importance. The dense vertical accumulation around zero reflects a large number of irrelevant or unstable features, whose contribution varies across random seeds. The black box on the left is just a visualization fail (it couldn’t fit at the top of the zero peak).

Further insight is provided by the bee-swarm and aggregated bar plots in Figures 8 and 9. Both consistently identify atm_580 as the most important and stable feature. Nevertheless, the classifier remains highly distributed, relying on the cumulative contribution of many features. This can be interpreted in light of brain physiology, where activity is led more by integration of different regions rather than segregation (which is also the reason why dimensionality reduction was not considered). The bee-swarm plot offers a global view of feature effects, where positive SHAP values increase the probability of stroke classification, while negative values decrease it.

The bar plot ranks features by mean absolute SHAP value, highlighting that although top-ranked features have the largest individual impact, the majority of the model’s predictive power arises from the remaining features. Finally, single-seed SHAP analyses (Fig. 7) allow interpretation of feature directionality, and mapping features back to the brain atlas enables anatomical interpretation of the most relevant cortical regions (Table IV).

D. Clinical application

Given the superior performance of models trained on ATM features, NA transitions appeared to be highly predictive of stroke. With further validation, this approach could support clinical prognosis rather than diagnosis. Once optimal features are identified, ATMs could be used to find target regions for neuromodulation protocols or in regression models to predict motor and neuropsychological recovery at different post-stroke time points, leveraging the outcomes already available in the TiMeS dataset and enabling more personalized therapeutic strategies.

IX. CONCLUSIONS, LIMITATIONS, AND FUTURE DEVELOPMENTS

In conclusion, this project identified the most predictive stroke-EEG features, showing that aperiodic brain activity captured by neuronal avalanches provides richer neuropathological information than traditional connectivity measures. This finding suggests a shift toward more informative and predictive EEG feature representations.

A key limitation is the small dataset size, which constrains model performance and generalizability. Larger and more diverse datasets would likely improve robustness, reduce performance variability, and yield more reliable results.

Future work will focus on using the most informative features (ATMs) to train regression models for predicting motor and neuropsychological recovery, supporting personalized rehabilitation strategies and improved clinical care.

X. ETHICAL IMPLICATIONS

For our project, the main ethical risk identified is misclassification if the model were to be used for diagnostic purposes. This risk is well documented in the medical AI literature, where inappropriate reliance on predictive models has been shown to lead to patient harm when models are insufficiently validated or applied outside their intended scope [9]. This could become potentially dangerous for patients, both in the case of healthy individuals and stroke patients. Over-trust in a model trained on a limited dataset may primarily harm vulnerable patients (stroke patients), but it can also affect clinicians who might rely on the output. The risk could be highly significant: false negatives could lead to inappropriate treatment decisions or delays in care delivery, whereas false positives could result in unnecessary anxiety and additional follow-up procedures for healthy individuals. Furthermore, over-reliance on the tool could introduce automation bias and shifting decisions away from clinical judgment. Overall, the severity is potentially high because misuse may have direct clinical consequences, while the likelihood is moderate because the current pipeline is research-oriented.

This risk was evaluated through the model performance metrics and the characteristics of the dataset, in line with established recommendations for assessing medical machine learning models [10]. The dataset is imbalanced (54 stroke versus 14 healthy controls) and restricted to T1 recordings (approximately one week post-stroke), which limits generalization. Models were assessed using balanced accuracy with an 80/20 stratified split, 3-fold cross-validation, and 50 runs with different random states. This procedure revealed large standard deviations, largely attributable to the small test set, that include roughly only 14 subjects. For the ATM-based SVM model, the balanced accuracy was approximately $74\% \pm 13\text{--}15\%$, further indicating uncertainty across splits. To take this risk into account, several mitigation strategies were implemented. Balanced accuracy was adopted as the primary metric and resampling was applied after the train and test split to avoid test leakage, together with cross validation to obtain more robust estimates. Furthermore, the work was explicitly framed as research and broader data and external validation were identified as necessary prerequisites before any clinical application. Nevertheless, despite these attempts to reduce misclassifications, the restricted dataset size, the absence of an independent external validation set, and the inability to perform a rigorous subgroup analysis remain the main barriers.

XI. APPENDIX

A. Correspondence between numbers and name of regions

Correspondence between the numbers of regions analysed in the SHAP analysis and the name of the regions.

TABLE V: ROI Index to Anatomical Name Mapping

Index	Anatomical Region Name
1	Caudal Anterior Cingulate L
2	Caudal Anterior Cingulate R
10	Fusiform R
14	Inferior Temporal R
19	Lateral Occipital L
23	Lingual L
29	Paracentral L
32	Parahippocampal R
42	Post Central R
43	Posterior Cingulate L

TABLE VI: Full List of Regions of Interest (ROIs) by Index and Anatomical Name

Index	Anatomical Region	Index	Anatomical Region
1	caudal anterior cingulate L	32	parahippocampal R
2	caudal anterior cingulate R	33	pars opercularis L
3	caudal middle frontal L	34	pars opercularis R
4	caudal middle frontal R	35	pars orbitalis L
5	cuneus L	36	pars orbitalis R
6	cuneus R	37	pars triangularis L
7	entorhinal L	38	pars triangularis R
8	entorhinal R	39	pericalcarine L
9	fusiform L	40	pericalcarine R
10	fusiform R	41	postcentral L
11	inferior parietal L	42	postcentral R
12	inferior parietal R	43	posteriorcingulate L
13	inferior temporal L	44	posteriorcingulate R
14	inferior temporal R	45	precentral L
15	insula L	46	precentral R
16	insula R	47	precuneus L
17	isthmus cingulate L	48	precuneus R
18	isthmus cingulate R	49	rostralanteriorcingulate L
19	lateral occipital L	50	rostralanteriorcingulate R
20	lateral occipital R	51	rostral middle frontal L
21	lateral orbitofrontal L	52	rostral middle frontal R
22	lateral orbitofrontal R	53	superior frontal L
23	lingual L	54	superior frontal R
24	lingual R	55	superior parietal L
25	medial orbitofrontal L	56	superior parietal R
26	medial orbitofrontal R	57	superiortemporal L
27	middle temporal L	58	superiortemporal R
28	middle temporal R	59	supramarginal L
29	paracentral L	60	supramarginal R
30	paracentral R	61	transverse temporal L
31	parahippocampal L	62	transverse temporal R

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